

Activity trace — accelerating matched SHD learning

ar073 · 19 July 2026 · Draft

Checkpoint CP-001

The next exploratory night is registered before any exp070 training result. This checkpoint has immutable private source SHA-256 prefix 5189910c09c9. Its checkpoint time is 2026-07-19 06:49:09.368 UTC, and it has no predecessor within this night. The raw snapshot is held read-only outside the repository. Hidden reasoning, tool payloads, injected environment material, credentials, private paths, addresses, and infrastructure identifiers are excluded. The visible attachment path is explicitly redacted; the request itself is reproduced verbatim.

Decision and action ledger

1. Pull request 52 is open and draft on the requested existing branch. Main is unchanged.
2. Exp069's complete epoch histories, firing diagnostics, gradients, rasters, split provenance, billing, and official-test prohibition were inspected.
3. A development-data-only audit rejected observation-window truncation as the likely bottleneck. The first registered candidate changes only the matched temporal resolution from 1 ms to 2 ms.
4. The ordered ladder, first-passing selection rule, promotion threshold, forty-epoch success criterion, kill criteria, one-seed policy, locked model comparison, and fresh paid-compute gate are fixed in ar072.
5. No exp070 runner, training result, cloud job, official-test access, or spend exists at this checkpoint. Pending work is the candidate-1 implementation, focused checks, local smoke, cost estimate, and paid-compute authority gate.

Privacy-sanitized visible transcript

User

2026-07-19 06:44:12.361 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Files mentioned by the user:

/goal Design and execute the next exploratory SHD experiment on PR #52, aimed a...: (PRIVATE ATTACHMENT PATH REDACTED)

The attached pasted text file(s) contain the user's request. Read and act on that content.

My request for Codex:

Attached request

/goal Design and execute the next exploratory SHD experiment on PR #52, aimed at materially improving the matched COBA/PING baseline while preserving a fair scientific comparison.

Context:

- Continue from exp069 on branch night/spiking-heidelberg-digits/ar070.
- exp069 trained for 80 epochs and reached 43.87% held-out validation accuracy for COBA and 48.04% for PING.
- The additional epochs provided insufficient benefit. Use much shorter runs while developing, then return to a 40-epoch matched run after selecting a convincing configuration.
- One seed is sufficient for this exploratory phase. Do not introduce multiple seeds until there is a claim worth defending.
- Never access or tune against the official SHD test set during exploration.

Scientific objective:

Identify a mechanistically justified change that substantially improves SHD learning—ideally improving both cells while revealing whether PING offers an advantage over COBA.

Investigate likely bottlenecks in this order:

1. input temporal representation/resolution and shared input scaling;
2. initialization and optimization settings;
3. inhibitory recruitment, firing-rate balance, and gradient flow;
4. existing or carefully added neuron/synapse time-constant mechanisms;
5. only if justified by the diagnostics, a new matched readout structure.

Locked comparison:

- Preserve Dale constraints.
- Keep model capacity, hidden population sizes, E/I counts, connectivity, and recurrence structure fixed.
- COBA uses no voltage-gradient dampening.
- PING uses voltage-gradient dampening 1000.
- Keep dataset split, checkpoint-selection rule, batching, and all non-cell-specific settings matched.
- Any new mechanism, including a readout change, must be applied identically to COBA and PING unless the difference is intrinsic to their registered cell definitions.
- Do not change the scientific design after seeing results without recording the result and obtaining approval for the revised design.

Readout and tools/snn authority:

- Do not casually search over losses or readouts.
- Begin with exp069's existing readout and loss.
- A new readout structure is allowed only when the diagnostics provide a clear scientific justification and it is pre-registered before its results are observed.
- Keep the classification objective matched and avoid introducing a readout that obscures the recurrent-

cell comparison.

- You are authorized to make careful edits or additions to tools/snn when needed, including support for a justified new readout structure or neuron/synapse mechanism.
- Preserve backward compatibility, keep changes narrowly scoped, add focused tests, and document the engine commit anywhere it affects scientific interpretation.
- Stop and ask if the required engine change becomes broad, invasive, or changes the meaning of the COBA/PING comparison.

Protocol:

1. Inspect exp069's implementation, artifacts, learning curves, firing diagnostics, rasters, and epoch-by-epoch behavior. Diagnose the most likely bottleneck before changing anything.
2. Create and lock a timestamped mandate in the next arNNN article. Pre-register the hypothesis, allowed intervention sequence, selection rule, success criterion, and kill criterion.
3. Run a one-seed local smoke stage first.
4. Iterate using very short registered runs, preferably about 5 epochs and no more than 10 unless justified. Use only the held-out validation split.
5. Use a small sequential set of scientifically motivated interventions, normally changing one factor at a time. Do not conduct an unconstrained hyperparameter sweep.
6. Record every meaningful attempt, including null, failed, and killed attempts, before proceeding.
7. Promote a configuration only if its short-run curves, losses, gradients, and firing diagnostics provide credible evidence that it should outperform the exp069 trajectory.
8. Freeze the promoted configuration and run matched COBA and PING training for 40 epochs.
9. Compare the new run against exp069 at matched epochs and against exp069's final 80-epoch checkpoint. Do not claim improvement through selective checkpoint reporting.
10. Do not use the official SHD test set. Stop at the held-out-validation human-review gate.

Compute:

- Run focused local checks first.
- Estimate cloud runtime and cost before requesting paid-compute authorization.
- Do not assume an earlier RunPod budget applies to this experiment.
- If authorized, use at most one pod per cell, respect the approved ceiling, and reap every pod immediately.
- Do not run the full test suite on the 4 GB Hetzner host. Use focused lightweight tests and demolab build.

Deliverables:

- Implement the experiment through experiments/expNNN.py and narrowly scoped tools/snn changes where justified.
- Produce artifacts/data/expNNN with complete provenance, numbers.json, reproducer, learning curves, firing-rate and gradient diagnostics, and matched input/E/I rasters.
- Create a cold-readable writings/expNNN.typ whose numbers and figures are sourced from recorded artifacts.
- Maintain timestamped scientific records in the relevant arNNN article: decisions, hypotheses, actions, failures, anomalies, costs, commits, results, conclusions, and pending work.
- Continue the existing research activity-log article with meaningful timestamped, hash-linked checkpoints.

- Make focused commits for implementation, engine changes, and every meaningful scientific attempt, including killed attempts.
- Build and validate the Demolab publication.
- Push the existing branch and update PR #52 with useful milestone comments.
- Do not merge the PR or modify main.
- Finish at the human-review gate with a concise evidence summary, exact compute spend, validation performed, rendered article/report links, relevant commit links, and the PR link.

Assistant

2026-07-19 06:44:16.949 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll load the lab runbook and the attached `/goal`, then verify the current branch/PR and resume from the recorded exp069 state. I'll also use the GitHub publishing workflow when the experiment reaches a commit/push milestone.

Assistant

2026-07-19 06:44:20.058 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The direct `demolab` command is not installed on this host, so I'm using the repository-prescribed `uvx demolab-cli docs` fallback before touching the experiment.

Assistant

2026-07-19 06:44:24.580 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

This is an AUTORESEARCH continuation, so I'm loading that runbook and its programme rules, plus the attached mandate and GitHub commit/push instructions. I won't start paid compute until I've reconciled the registered design with the current branch and its existing records.

Assistant

2026-07-19 06:45:10.455 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The programme is in "changes requested" state: you have explicitly approved a new exploratory night on the existing PR branch. I'm treating the attached goal as the mandate-gate approval, preserving its scientific terms verbatim, and keeping `main` untouched as requested; the branch commit will remain the pre-result anchor.

Assistant

2026-07-19 06:45:44.949 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The exp069 evidence is intact locally, including full epoch histories and rasters. I'm now quantifying its early-epoch bottleneck, checking the next free IDs, and locating the existing private-trace custody pattern so the new night begins with a complete, sanitized checkpoint rather than a reconstructed summary.

Assistant

2026-07-19 06:47:51.951 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first input audit is testing two concrete failure modes on development data only: how much speech exp069 discarded after 1.0 seconds, and how much temporal information coarser bins would collapse. This precedes intervention selection and does not touch the official test split.

Assistant

2026-07-19 06:49:07.003 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The diagnosis points to temporal discretization as the first clean intervention. Exp069 discarded only 0.017% of development events after 1 second, so extending duration is unlikely to move accuracy. A 2 ms bin collapses only 0.094% of events in a deterministic 1/8 development sample, while halving backpropagation depth; 4 ms already collapses 2.04% and is coarse relative to the 2 ms AMPA time constant. I'm preregistering 2 ms first, before observing any new training result.